COLLEGE CODE - 7177

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PHASE 2 - SUBMISSION

EARTHQUAKE PREDICTION ANALYSIS

Earthquake Prediction is a crucial area of research in seismology and geophysics . In this project,We aim to develop data-driven earthquake prediction model using python and kaggle dataset containing historical earthquake data . The primary goal is to analyse patterns and build machine learning model that can provide probabilistic earthquake prediction . Additionally , we are using the advanced techniques such as hyperparameter tuning and feature engineering to improve the prediction model's performance.

Dataset Extraction :

https://www.kaggle.com/datasets/usgs/earthquake-database

Major Input Columns used in dataset:

- Date
- Time
- Latitude
- Longitude
- Depth

Output:

Using Date is a normal thing ,it is not required for the EarthQuake prediction , EarthQuake can be predicted by using the data such as TimeStamp , Latitude and Longitude are required for the prediction which gives the output as magnitude and depth in which if the magnitude is **greater than 5.5** is detected as EarthQuake

Innovative Techniques:

To improve prediction system accuracy and robustness we use ensemble method from sklearn.ensemble , we import RandomForestRegressor to predict the system's accuracy . we build the neural network to fit the data for training set. Neural Network consists of three Dense layer with each 16, 16, 2 nodes and relu, relu and softmax as activation function from the keras.models ,we import Sequential and keras.layers , we import Dense .

Libraries to be used and Downloaded :

- numpy - for scientific computing
- pandas - for data manipulation and analysis
- matplotlib – for data visualization2
- folium – for interative map visualization
- scikit-learn – for machine learning
- keras – for deep learning
- RandomForestRegressor (sklearn.ensemble) – for ensemble methods

Train and Test methods :

from library sklearn.cross_validation ,we import train_test_split which is used to train and test the model ,in which Test Dataset contains 20% and Training Dataset contains 80% and after checking,we train the data using RandomForestRegressor which is used to predict the accuracy of the training model by testing using 20% dataset .

Predictive Measures to improve the accuracy :

To improve the accuracy of the predictive model , we use deeplearning model , we built the neural network to fit data for training set . Neural Network consists of three Dense layer with each 16, 16, 2 nodes and relu, relu and softmax as activation function. we use keras library for that and we use Tensorflow as backend.